MIL-STD-639 (Ord)
10 July 1958

SUPERSEDING
Ordnance Engineering
Standard URAX6
7 March 1958

MILITARY STANDARD

SURFACE ROUGHNESS VALUES PREFERRED FOR ARMY ORDNANCE MATERIEL

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DEPARTMENT OF THE ARMY
ORDNANCE CORPS
WASHINGTON 25, D. C.

SURFACE ROUGHNESS VALUES PREFERRED FOR ARMY ORDNANCE MATERIEL

MIL-STD-639 (Ord)

1. This standard has been approved by the Department of the Army, and is published as a guide for selecting surface roughness values preferred for Ordnance materiel.

2. Recommended corrections, additions or deletions should be addressed to Office, Chief of Ordnance, Department of the Army, Washington 25, D. C.

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1. SCOPE

1.1 General - This standard establishes minimum surface roughness values for application to various types of Ordnance materiel.

1.2 Intended use - The surface roughness values tabulated in this standard are intended for use by engineers, designers and draftsmen as guides in the selection of proper surface roughness values. The standard surface roughness comparison specimens illustrated are intended for use by inspectors and shop personnel, to check finished surfaces by sight or touch against the standard specimen finish.

2. REFERENCED DOCUMENTS

MIL-STD-10 - Surface Roughness Waviness and Lay

3. DEFINITIONS

The definitions of MIL-STD-10 apply to this standard.

4. REQUIREMENTS

4.1 Assignment of values - All detail drawings of Ordnance materiel to which surface roughness values apply shall have the roughest surface consistent with the intended use of the item. Acceptance of finer surfaces than those specified is permissible without further authorization except when such results in increased cost to the government.

4.1.1 Natural surfaces - Natural finishes (castings, forgings etc) shall be within the values indicated. These surface roughness are not usually indicated on drawings as they are dependent on the finish of the mold, or die.

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4.2 Application - Surface roughness values for Ordnance material shall be selected, when applicable from the accompanying tables.

4.2.1 Use of the tables - The material in tables I to VIII is so arranged that various classes of the same type of surface are grouped together for ease of selection. Table IX, in the appendix, is a reference table showing the range of finishes obtainable with different industrial processes, the range of surface roughness to be expected on some of the more common metals, and a cross reference between the old "f" system of surface roughness designation and the arithmetic mean or arithmetic average system.

4.2.1.1 Surface roughness numbers - The surface roughness numbers for each application listed are the arithmetic average (AA) values expressed in micro-inches (See MIL-STD-10). The AA value for each application listed is the roughest finish acceptable for that type surface and is generally the most economical to use.

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TABLE I

NON MATING SURFACES

AA Roughness Height Ratings	500	250	125	63	32	16	8	4
A Clearance Holes		X						
B Clearances & Reliefs								
1. Small			X					
2. Medium or Large	X							
C Cutoff length Surfaces-sheared, sawed, etc.	X							
D Datum Surfaces								
1. Less than .001" tolerance				X				
2. Tolerance of .001"			X					
E Nuts, Bolt & Screw Heads, unthreaded shanks								
1. Finished (machined) bolts, screws			X					
2. Unfinished bolts		X						
F Ends of bolts, pins, screws & studs		X						
G Screwdriver & Wrench slots		X						
H Chamfers, Radii and Undercuts			X					
I Handles			X					
J Tool Runout-thread relief			X					
K Exterior Surfaces								
1. Housings cast	X							
2. Housings machined			X					
3. Guns through 30 MM			X					
4. Guns over 30 MM to 16"		X						
5. Painted Surfaces, guns 75MM to 16"	X							
6. Breech Blocks				X				
L Breech Mechanisms								
1. Guns through 30 MM				X				
2. Guns larger than 30MM to 125MM			X					
3. Guns 125MM to 16"		X						

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TABLE II

MATING OR CONTACT SURFACES - STATIONARY

AA Roughness Height Ratings	500	250	125	63	32	16	8	4
A Centralizing or locating surfaces			X					
B Clamping or mounting surfaces				X				
C Housing, bracket and pedestal-pads (Base Surfaces)			X					
D Surfaces for copper gaskets & gasket seats					X			
E Surfaces for soft, flat gaskets			X					
F Gasket surfaces (minimum surface contact)				X				
G Grooves for Injection Seals			X					
H Surfaces for "0" Rings					X			
I Grooves for Snap Rings			X					
J Counterbored Surfaces								
1. Over 3/4 diam.		X						
2. 3/4 diam. & less			X					
K Countersunk Surfaces			X					
L Spotfaced Surfaces								
1. Over 3/4 diam.		X						
2. 3/4 diam. & less			X					
M Dowel Pin holes and taper pin holes					X			
N Parts of Breech mechanism				X				
O Inside diam. of pinned hubs, collars spacers				X				
P Lens, prism & mirror mounting surfaces				X				
R Spring Seat Surfaces			X					
S Shafts & bores for ball bearings								
1. Up to 2" diam.					X			
2. Over 2" diam.				X				
T Shoulder faces for shafts and Housings (Ball Races)			X					
U Surfaces contacting packing in Glands and Retainers				X				

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TABLE III

MATING OR BEARING SURFACES - SLIDING

AA Roughness Height Ratings	500	250	125	63	32	16	8	4
A Gear teeth and screw threads								
1. DP 10 or smaller								
a. General				X				
b. Precision					X			
2. Coarser than 10 DP				X				
3. Heavy loads					X			
4. Worms						X		
5. Worm Gears								
a. General					X			
b. Precision (lapped)						X		
c. For heavy loads					X			
6. Teeth of ratchets and pawls			X					
7. Spline teeth				X				
8. Screw threads								
a. Chased		X						
b. Die or tap cut			X					
c. Milled								
1. 10th'ds per inch or smaller				X				
2. Larger than 10th'ds per inch			X					
d. Ground th'ds & breech threads for guns						X		
e. Rolled th'ds							X	
B Gibs and Ways					X			
C Sliding plates					X			
D Sliding plate guides						X		
E Slip Clutch Surfaces								
1. Metal to Metal				X				
2. Metal to non-metal					X			
F Slip ring Surfaces				X				
G Valve stems and guide bushings						X		
H Cylinder bores, pistons & piston rods						X		
I Surfaces of fluid seals								X
J Valve Seats							X	
K Bearing Surfaces bolts, nuts, screw heads			X					

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TABLE III (Cont'd)

MATING OR BEARING SURFACES - SLIDING

AA Roughness Height Ratings	500	250	125	63	32	16	8	4
L Cam surfaces and followers								
1. Three dimensional						X		
2. Groove								
a. General					X			
b. Precision						X		
3. Flat or disc lobe								
a. General					X			
b. Precision						X		
4. Throwout type				X				
M Locking plungers (round or square) and holes				X				
N Keys and keyways					X			
O Breech & firing mechanisms of cannons						X		
P Parts sliding in Packings						X		
R Dynamic "0" Ring seal surfaces								X
S Dynamic "T" Seal (Machined Finish - No Abrasive)						X		
T Recoil Mechanisms & Equilibrators								
1. Anti Friction Metal						X		
2. Copper Rings					X			
3. Silver Rings						X		
4. Control rods-bronze Buffer ends				X				
5. Control rods;steel-control diameter					X			
6. Internal Bronze Surfaces				X				
U Propellent Valves Shafts						X		
V Rifling in Cannon Barrels								
1. Lands								
a. Cannon over 30MM up to 75MM							X	
b. Cannon 75MM up						X		
2. Grooves								
a. Cannon over 30MM up to 75MM						X		
b. Cannon 75MM up					X			
W Barrel chambers, lands & grooves								
1. Guns through 30 MM					X			

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TABLE IV

MATING OR BEARING SURFACES - ROTATING

AA Roughness Height Ratings	500	250	125	63	32	16	8	4
A Crankpins							X	
B Pivot Holes - Pivot Pins					X			
C Bearings - Ball Track							X	
D Bearings Sleeve type (Shaft OD - Bearing ID)								
1. General					X			
2. Precision						X		
E Shaft OD used with Jewel bearing								X
F Shaft OD used with oil seal or "0" Ring						X		
G Piston Pins								X
H Friction Differential faces							X	
I Variable Speed Drives - Cone, Disc and Cylinder faces							X	
J Hub, collar and shaft face bearing Surfaces								
1. General				X				
2. Precision					X			
K Pressure lubricated Bearings								X
M Propellent Blades								X

TABLE V

INTERFERENCE FITS

AA Roughness Height Ratings	500	250	125	63	32	16	8	4
A Push Fit					X			
B Keys and Keyways				X				
C Drive and Press Fits								
1. Holes & shafts to 2" Diam.					X			
2. Holes & shafts over 2" Diam.				X				

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TABLE VI

AMMUNITION COMPONENTS

	AA Roughness Height Ratings,	500	250	125	63	32	16	8	4
A	Shell, rifle bore, 155MM and up (except as noted below)	X							
	1. Cavity, cannister, smoke, B.E.; All Bourrelets		X						
B	Shell, rifle bore, 57MM to 120MM inc.		X						
C	Shell, rifle bore, 20MM to 40MM inc., except base			X					
D	Shell, rifle bore, 20MM to 40MM inc., (base only)		X						
E	Shell, smooth bore, mortar (except Bourrelet)	X							
F	Shell, smooth bore, mortar (Bourrelet)			X					
G	Shot, 57MM to 8" inc.		X						
H	Shot, 20MM to 40MM inc.			X					
I	Cartridge, drill, 57MM to 105MM inc.	X							
J	Cartridge, drill, 20MM to 40MM inc.			X					
K	Projectile, dummy (except mating surfaces)	X							
M	Projectile, dummy (mating surfaces)			X					
N	Tracer cavities-all projectiles		X						
O	Tracer body, M3			X					
P	Burster castings & adapters (projectiles)	X							
R	Fuses, primers, boosters, bursters, grenades, mines, pyrotechnics (only surfaces in contact with loose or pelleted explosive)			X					
S	Fuses, primers, boosters, bursters, grenades, mines, pyrotechnics (sliding surfaces only)				X				
T	Fuses, primers, bursters, grenades, mines, pyrotechnics (all surfaces other than above)		X						
U	Cartridge cases, except primer pocket		X						

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TABLE VI (Cont'd)
AMMUNITION COMPONENTS

AA Roughness Height Ratings	500	250	125	63	32	16	8	4
V Cartridge cases, primer pocket only				X				
W Bomb bodies including metal components for them	X							
X Chemical bombs, nose openings & adapters only where force fit is required. (Surfaces to be welded or brazed not included)				X				
Y Warheads		X						

TABLE VII
GAGE SURFACES

AA Roughness Height Ratings	250	125	63	32	16	8	4
Component contacting surfaces presenting line or point contact							X
Critical slide and bearing fits							X
Less critical slide fits (flush pins)*						X	
Feeler or sighting surfaces					X		
Precision located snug, push, drive or press fits*					X		
Free or running fits				X			
Reference surfaces for measuring purposes					X		
Non precision located snug, push, drive or press fits*			X				
Overall cleanup finishes (hand gages)			X				
Overall cleanup finishes (castings or other unimportant machined surfaces)	X						

*The class fit shall be considered when specifying the required finish.

NOTE: Values for natural finishes (castings, forgings, etc.) shall not be specified on gage drawings.

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TABLE VIII

SURFACE ROUGHNESS VALUES FOR GAGES ACCORDING TO SIZE & TOLERANCE

SIZE RANGE		Apply ✓4 to all gaging tolerances between these values	
Above	To & Incl.		
.029	.825	.00003	.00009
.825	1.510	.00004	.00011
1.510	2.510	.00006	.00014
2.510	4.510	.00007	.00017
4.510	6.510	.00009	.00022
6.510	9.010	.00012	.00028
9.010	12.010	.00015	.00035

Apply ✓2 to all gaging tolerances below these values

Apply ✓8 to all gaging tolerances above these values

NOTE: The above surface finish values apply to all gaging surfaces of the following:

a. All types of plug and ring gages (except thread gages).
b. Snap gages (except adjustable types).
c. Plug type check or limit gages.
d. Fixture gaging elements subject to wear.

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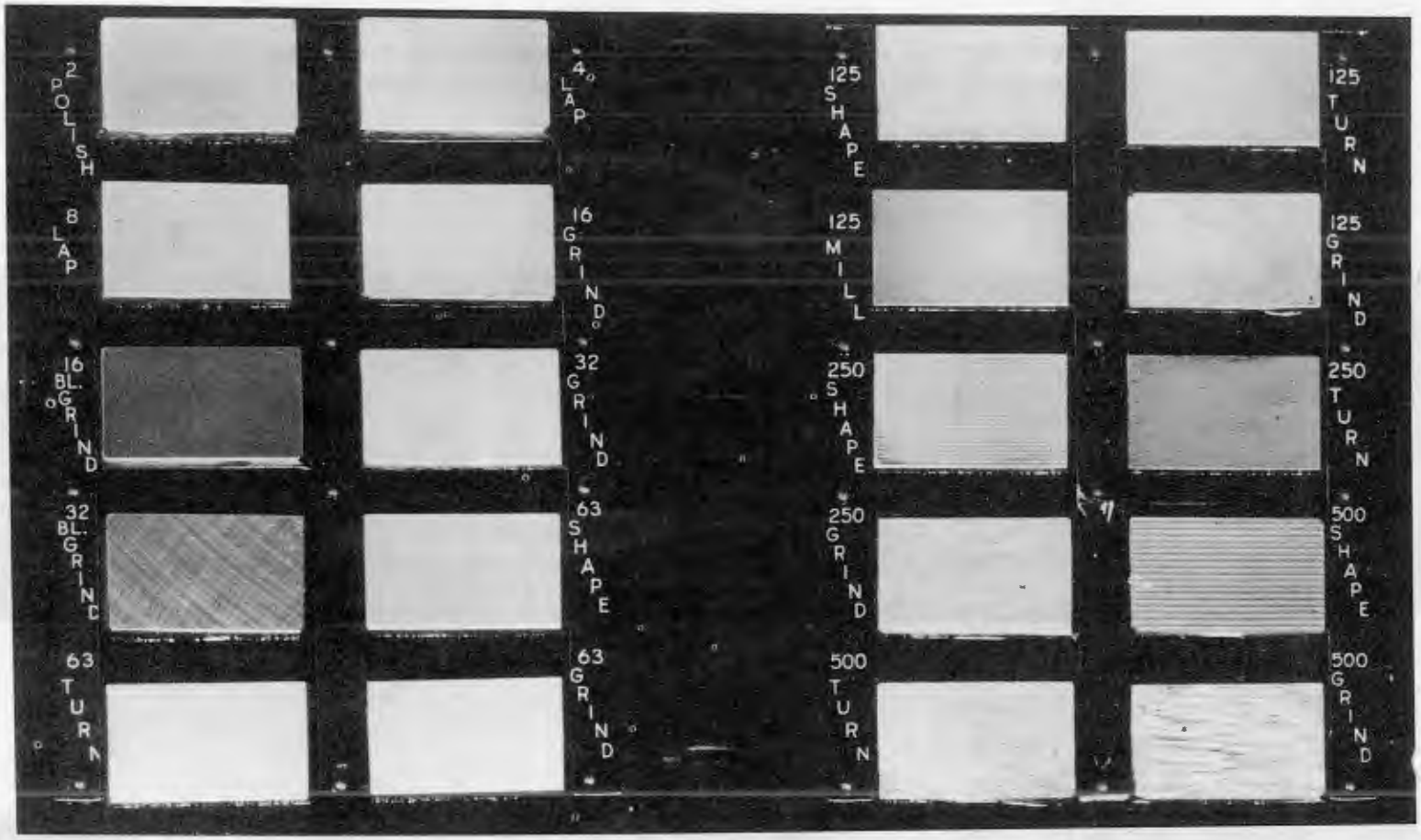


FIGURE 1

Specimens of Machine Finishes for Metal Surfaces Type I

NOTE: These specimens may be obtained through the Commanding General, Frankford Arsenal, Philadelphia 37, Pa., Attn: ORDBA-GC

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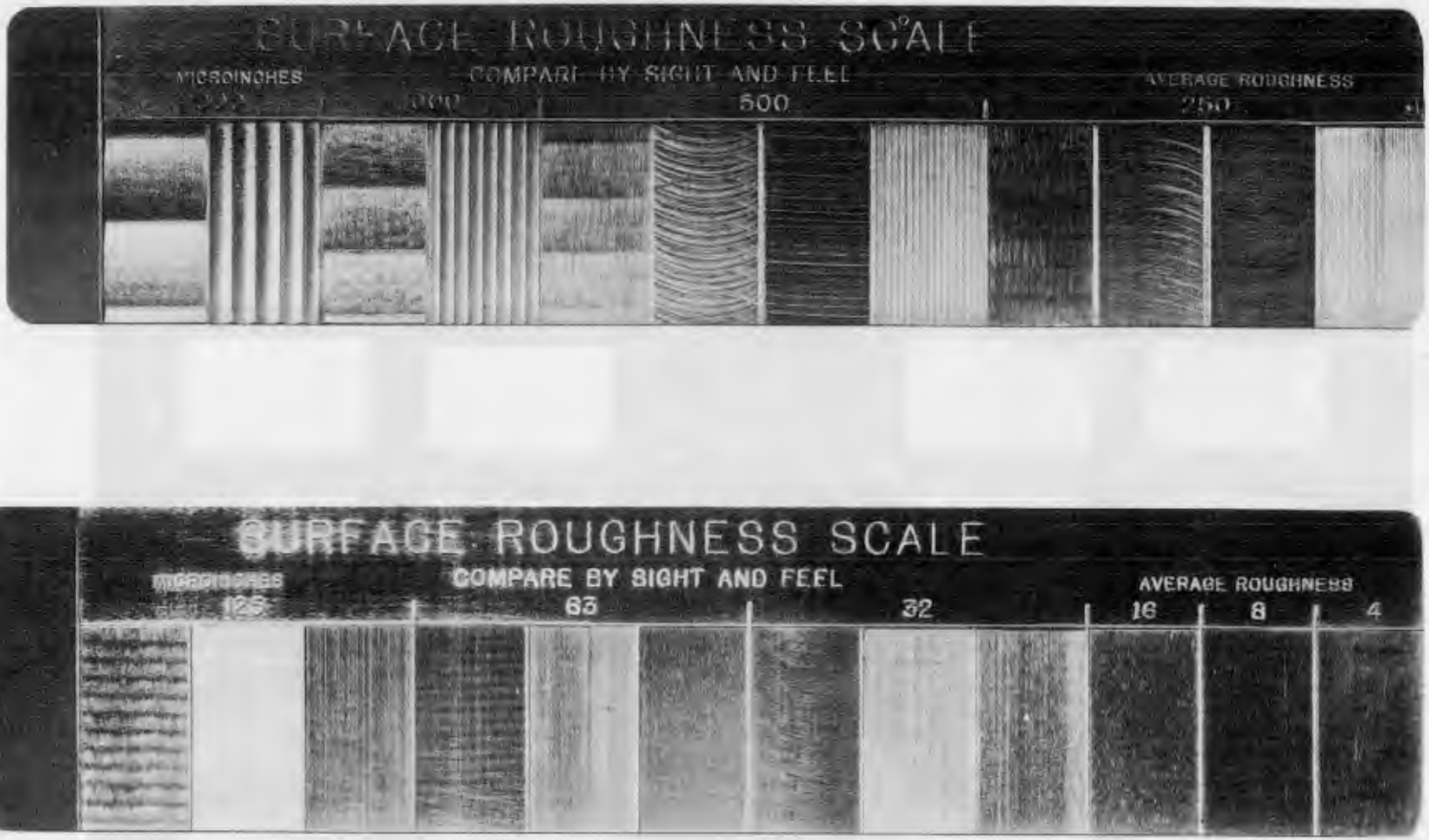


FIGURE 2

Specimens of Machine Finishes for Metal Surfaces Type II

NOTE: These specimens may be obtained through the Commanding General, Frankford Arsenal, Philadelphia 37, Pa., Attn: ORDBA-GC

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APPENDIX

There are no mandatory requirements in this appendix

10. FACTORS AFFECTING SURFACE ROUGHNESS - The primary reason for surface control is to improve performance, increase effective service life or reduce cost. The final choice may depend on such factors as speed, type of loading, direction of motion, physical characteristics of both materials in contact, type and amount of lubrication, operating temperatures and in some cases, appearance. In general the smoother the surfaces, when in moving contact with adequate lubrication, the more efficiently they will perform, the longer they will last and the less maintenance they will require. If however the surfaces are too smooth they can cause lubrication problems, and without adequate lubrication may freeze due to molecular adhesion. Experience has shown that the surface values in this standard are optimum for the use listed.

10.1 Use of surface control numbers - Surface roughness values should be used only on surfaces requiring control. All other surfaces should be left to the finish provided by the process which formed them.

20. CONVERSION FACTORS (RMS TO AA) - Since the adoption of the arithmetic average (AA) as the criterion for surface roughness, problems have developed regarding the change over, as the Root Mean Square (RMS) figure is about 11% rougher for the same surface. A surface tester calibrated to the RMS scale

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will therefore accept a surface rougher than intended unless it is re-calibrated to the AA system. If this is not practical, an 11% correction factor should be applied to the readings (see 5.2.4 of MIL-STD-10). This correction factor should be used only on critical surfaces as the 11% difference is less than the point to point variation over a machined surface and much less than the variation expected from piece to piece.

30. SURFACE ROUGHNESS COMPARISON SPECIMENS - Two types of surface roughness comparison specimens are available for either tactual or visual comparison with actual machined surfaces.

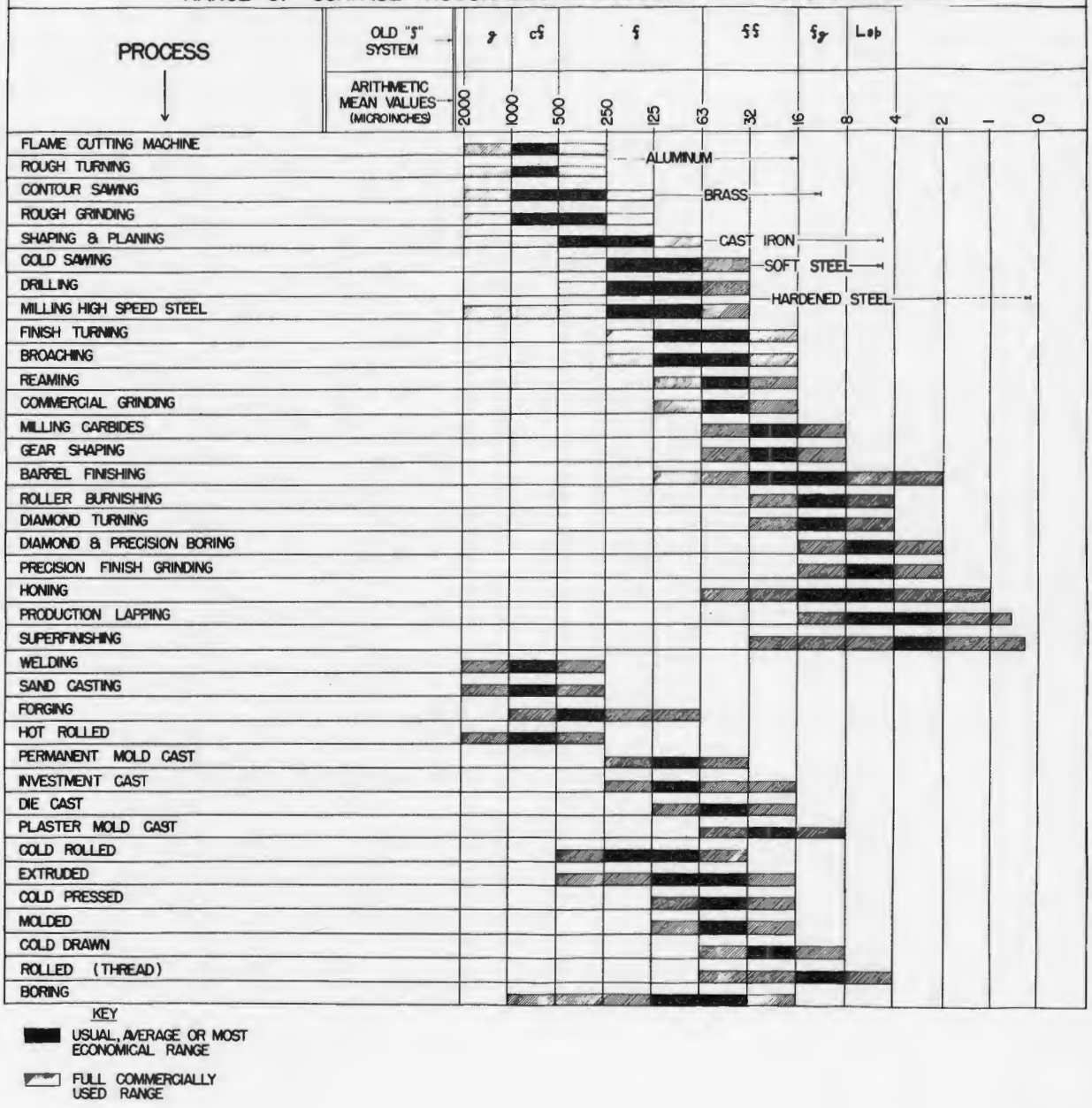
Type I are original type specimens (see Figure 1) Ord Stock No. 7949997

Type II are replica type specimens (see Figure 2) Ord Stock No. 7949998

30.1 Old type II & type III specimens - The former type II specimens are now listed as Type I (see Figure 1). The former type III specimens are now listed as type II (see Figure II). These older specimens are made to the RMS values but will remain in use unchanged as the difference in surface roughness between the RMS system and the AA system is not detectable by touch or vision.

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TABLE IX
RANGE OF SURFACE ROUGHNESS PRODUCED BY VARIOUS PROCESSES



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